



PHYSICS

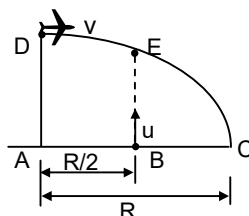
Max Marks: 60

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

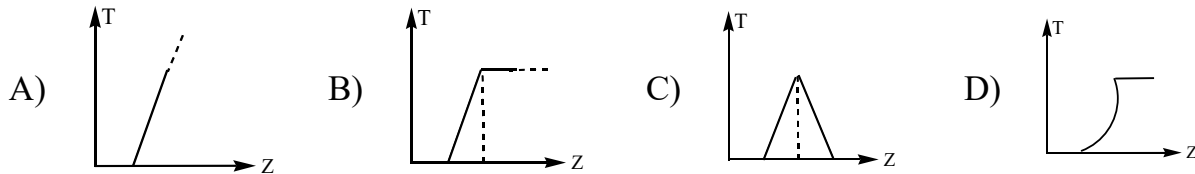
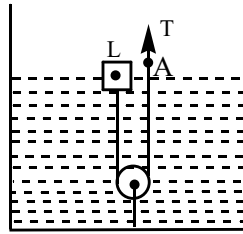
This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

- The pitch of a screw gauge is 0.5mm and there are 100 divisions on its circular scale. The instrument reads +2 divisions on circular scale when nothing is put in between jaws. In measuring diameter of a wire, there are 8 divisions on the main scale and 83rd division of circular scale coincides with reference line. The diameter of wire is
A) 4.425 mm B) 4.162 mm C) 8.405 mm D) 8.415 mm
- A bomb is dropped from D from an aeroplane moving horizontally with a speed v . At the same instance another bomb is vertically projected with a speed u from the point B. what should be u so that the two bombs collide at E while bomb from B is going up.



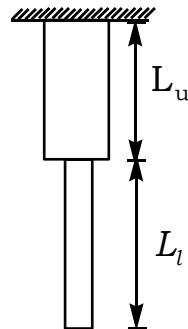
- A) v B) $v/2$ C) $\frac{gR}{v}$ D) $\frac{\sqrt{2}gR}{v}$
- A sphere moving with velocity v towards a wall strikes elastically with the wall moving towards the sphere with a velocity u . If the mass of the wall is infinitely large, the net work done by the wall on the ball during collision will be
A) $mu(u+v)$ B) $2mu(u+v)$ C) $2mv(u+v)$ D) $2m(u+v)$
 - A cubical wooden block of side L and relative density 0.5 is floating on the surface of water as shown in the figure. A string is attached to it through an ideal pulley fixed to the bottom of vessel as shown in figure. The graph of tension T in the string vs distance of the bottom of the block from the free surface of the water when the end A of the rope is slowly pulled up will be (block is moving very slowly and the vessel is very wider):



5. A composite heavy rope of two materials is suspended vertically from a high ceiling. The ratios of different quantities for upper to lower rope are:

Lengths $\frac{L_u}{L_l} = \frac{1}{2}$, Cross sectional areas $\frac{A_u}{A_l} = \frac{2}{1}$, Densities $\frac{d_u}{d_l} = \frac{2}{3}$.

What is the ratio of maximum stress in upper rope to that in lower rope?



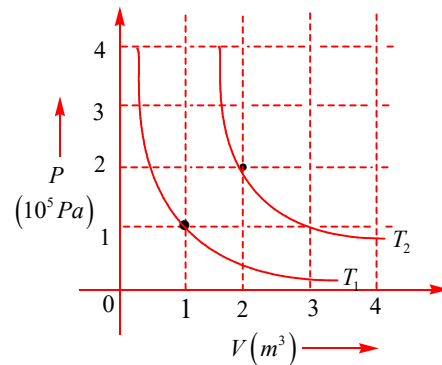
- A) $\frac{6}{5}$ B) $\frac{5}{6}$ C) $\frac{3}{2}$ D) $\frac{2}{3}$
6. When a floating object like a ship moves, it also has to move the water that it displaces. Due to this the 'effective mass' of the ship is increases by approximately one third of the mass of the ship. A boat of total mass M is moving at speed u and is accelerating at a rate a in a large lake. If P is the power delivered by the motor of the boat, then pick the most appropriate option without neglecting viscosity of water.

- A) $P < \frac{2}{3} Mau$ B) $P = Mau$
C) $P > \frac{4}{3} Mau$ D) The net force acting on the boat is $M\sqrt{g^2 + a^2}$

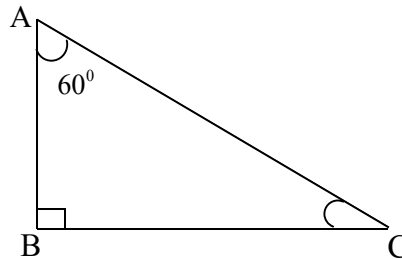
7. A glass capillary tube of inner radius r is immersed in water. Find the quantity of heat, that will be evolved when water rises in tube. The surface tension of water is T , angle of contact between water and glass is 0° and density of water is ρ .

A) $\frac{2\pi T^{3/2}}{\rho^2 g}$ B) $\frac{2\pi T^2}{\rho g}$ C) zero D) $\frac{\pi T^2}{\rho g}$

8. The following graph shows two isotherms for a fixed mass of an ideal gas. The ratio of r.m.s speed of the molecules at temperatures T_1 and T_2 is

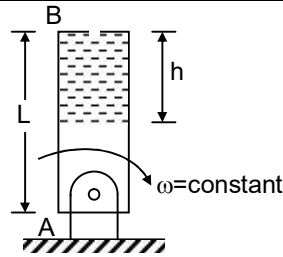


- A) $2\sqrt{2}$ B) $\sqrt{2}$ C) 2 D) 4
9. Three rods of the same cross section and made of the same material form the sides of a triangle ABC as shown. The points A and B are maintained at temperatures T and $\sqrt{2}T$ respectively in the steady state. Assuming that only heat conduction takes place, the temperature at point C is



A) $\left[\frac{2\sqrt{2} + \sqrt{3}}{2 + \sqrt{3}} \right] T$ B) $\left[\frac{3\sqrt{2}}{2 + \sqrt{3}} \right] T$ C) $\left[\frac{2}{\sqrt{3}} \right] T$ D) $\left[\frac{\sqrt{5}}{2} \right] T$

10. A tube of uniform cross section 'a' and length L is rotating with a constant angular speed ω in the vertical plane. End A is open to atmosphere. What is the angular speed required just to stop water coming out from a very small opening present at the closed end B when the hole is at the top most position. Atmospheric pressure is P_0 (pascal)



A) $\sqrt{\frac{2g}{(2L-h)}}$

B) $\sqrt{\frac{2(\rho hg - p_0)}{\rho h(2L-h)}}$

C) $\sqrt{\frac{2gh}{L^2 - h^2}}$

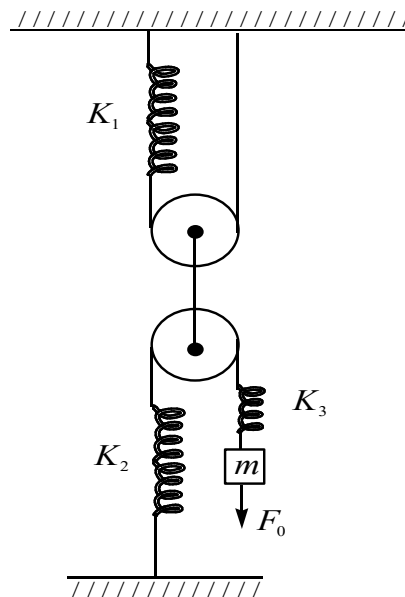
D) $\sqrt{\frac{2(\rho hg - p_0)}{\rho(L^2 - h^2)}}$

SECTION - II (PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 11 & 12

In the figure shown $k_1=200 \text{ N/m}$ $k_3=k_2=400 \text{ N/m}$, $m=2 \text{ kg}$ and $F_0=80 \text{ N}$. The system is at rest and in static equilibrium. ($g=10 \text{ m/s}^2$)



11. Elongation in spring k_1 is
 A) 0.1 m B) 0.5 m C) 0.25 m D) 0.20 m
12. Now the force F_0 is removed suddenly. Then displacement of the block from the given instant to its equilibrium position is
 A) 80 cm B) 60 cm C) 40 cm D) 50 cm

Paragraph for Question Nos. 13 & 14

A rigid body can be hinged about any axis parallel to Z –axis and passing through a point on the X-axis. When it is hinged about an axis passing through $(x,0,0)$ its moment of inertia is given by $I = 2x^2 - 12x + 27$; where x is in meter and I is in kg-m^2 .

13. The x-coordinate of the centre of mass of the body is:
 A) $x=2m$ B) $x=3m$ C) $x=0$ D) $x=1m$
14. The mass of the body is:
 A) 2kg B) 3 kg C) 1kg D) 4 kg

Paragraph for Question Nos. 15 & 16

A physics tutor launches a home-built model rocket straight up into the air. At $t = 0$, the rocket is at $y = 0$ with $u_{y(t=0)} = 0$. The acceleration of the rocket is given by

$$a_y = \begin{cases} -g + \alpha g - \beta t^4, & 0 < t < t_b \\ -g, & t > t_b \end{cases}$$

where $t_b = \left(\frac{\alpha g}{\beta}\right)^{1/4}$ is the time at which fuel burns out. α is a positive dimensionless number (> 1)

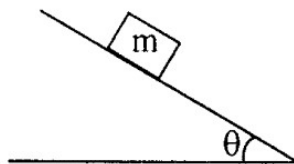
15. The expression for the velocity $v_y(t)$ valid at all times in the interval $0 < t < t_b$ is __
 A) $v_y = (\alpha - 1)gt + \frac{1}{5}\beta t^5$ B) $v_y = (\alpha - 1)gt - \frac{1}{5}\beta t^5$
 C) $v_y = (\alpha + 1)gt + \frac{1}{5}\beta t^5$ D) $v_y = (\alpha + 1)gt - \frac{1}{5}\beta t^5$
16. The time taken for rocket to reach its maximum height is
 A) $\frac{3}{5}\alpha t_b$ B) $\frac{4}{5}\alpha t_b$ C) $\frac{1}{5}\alpha t_b$ D) $\frac{2}{5}\alpha t_b$

SECTION - III
(Matching List Type)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

17. A block placed on a rough inclined plane angle of inclination θ of the plane as shown is gradually varied starting from zero. The coefficient of static friction and kinetic friction between the block and the plane is μ_s and μ_k respectively ($\mu_s > \mu_k$). Column-II shows the graph which necessarily contains θ taken on x-axis. Column-I represents the quantities taken on y-axis of column-II

**COLUMN-I**

P) Friction force between the block and plane

Q) Normal force between the block and plane

R) Total contact force between the block and plane

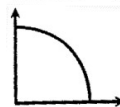
S) Acceleration of block

COLUMN-II

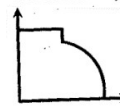
(1)



(2)



(3)



(4)



A) P-4, Q-3, R-2, S - 1

B) P-4, Q-2, R-3, S - 1

C) P-2, Q-4, R-1, S - 3

D) P-1, Q-2, R-3, S - 4

18. The magnitudes of gravitational field at distances r_1 and r_2 from the center of a uniform sphere of radius R and mass M are respectively I_1 and I_2 . The ratio I_1 / I_2 is given in the column II. Make correct matching

	Column – I		Column – II
P)	$r_1 > R$ and $r_2 > R$	1)	$\frac{r_1 r_2^2}{R^3}$
Q)	$r_1 < R$ and $r_2 < R$	2)	$\frac{R^3}{r_2 r_1^2}$
R)	$r_1 > R$ and $r_2 < R$	3)	$\frac{r_1}{r_2}$
S)	$r_1 < R$ and $r_2 > R$	4)	$\frac{r_2^2}{r_1^2}$

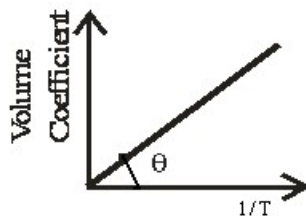
A) P-4, Q-3, R-2, S - 1

B) P-4, Q-2, R-3, S - 1

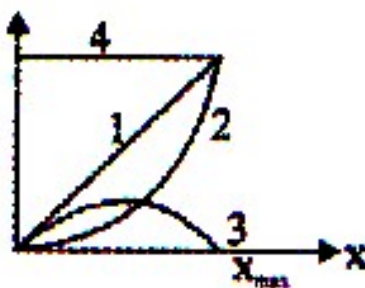
C) P-2, Q-4, R-1, S - 3

D) P-1, Q-2, R-3, S - 4

19. A monatomic ideal gas of 2 moles was found to be undergoing a thermodynamic polytropic process ($PV^x = \text{Constant}$ where x is a real number) whose volume coefficient is varying with temperature as shown. (Take $R = 25/3 \text{ J/mol/K}$)

**Column I****Column II**

- P) Work done by gas in joules for every 1 K rise in temperature if $\tan \theta = 3$. 1) 50
- Q) Internal energy rise of the gas in joules for every 1 K rise in temperature if $\tan \theta = 2$. 2) 25
- R) Heat absorbed by the gas in joules for every 1 K rise in temperature if $\tan \theta = 1$. 3) $125/3$
- S) Value of ' θ ' in degrees for which the internal energy rise of the gas for every 1K rise in temperature is 25 J. 4) can be any of the above values
- A) P-4, Q-3, R-2, S - 1 B) P-4, Q-2, R-3, S - 1
- C) P-2, Q-4, R-1, S - 3 D) P-1, Q-2, R-3, S - 4
20. A block suspended from a spring is released from rest when spring is unstretched. ' x ' represents stretch in spring. Select the appropriate graph from column-II taking quantities in column - I as y-axis.

**Column I** (Quantity on y-axis)**Column II**

- P) The KE of block 1) 1
- Q) The work done on the block by gravity 2) 2
- R) The magnitude of work done on the block by spring 3) 3
- S) The total mechanical energy of block - earth - spring system 4) 4
- A) P-4, Q-3, R-2, S - 1 B) P-4, Q-2, R-3, S - 1
- C) P-2, Q-4, R-1, S - 3 D) P-3, Q-1, R-2, S - 4

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

21. A compound H_2X with molar weight of 80g is dissolved in a solvent(solvent density is 0.4 g ml^{-1}). Assuming no change in volume upon dissolution, the molality of a 3.2 molar solution is
A) 2.5 B) 4.5 C) 6.0 D) 8.0
22. A partially dried clay mineral contains 8% water. The original sample contained 12% water and 45% silica. The % of silica in the partially dried sample is nearly:
A) 50% B) 49% C) 55% D) 47%
23. For each of the following pairs of E_2 reaction, select the one that occurs with larger rate constant
1) $CH_3CH_2CH_2Cl + (CH_3)_3CO^-$ or 2) $CH_3CH_2CH_2I + (CH_3)_3CO^-$
3) $CH_3CH(Br)CH_3 + CH_3O^-$ or 4) $CH_3CH(Br)CH_3 + CH_3S^-$
5) $PhCH_2CH(Br)C_2H_5 + OH^-$ or 6) $PhCH_2CH_2CH(Br)CH_3 + OH^-$
A) 2,4,6 B) 1,3,5 C) 2,3,5 D) 2,4,5
24. 1 litre flask contains nitrogen along with a drop or two of water at $40^\circ C$, the total pressure being 760 Torr. If all the contents are transferred to another flask of 0.5L at the same temperature. The total pressure of nitrogen setup in the second flask will be (aqueous tension at $40^\circ C = 55 \text{ torr}$)
A) 1410 Torr B) 1465 Torr C) 1520 Torr D) 1540 Torr
25. A mixture of $NH_3(g)$ and $N_2H_4(g)$ is placed in a sealed container at 300 K. The total pressure of the gaseous mixture is 0.5 atm. The container is heated to 1200 K at which both NH_3 and N_2H_4 decomposed completely according to the following equation .
$$2NH_3(g) \longrightarrow N_2(g) + 3H_2(g); \quad N_2H_4(g) \longrightarrow N_2(g) + 2H_2(g)$$

After decomposition is complete , the total pressure at 1200 K is found to be 4.5 atm. The mole % of N_2H_4 in the initial mixture at 300 K was
A) 20% B) 25% C) 50% D) 75%
26. 'n' mole of helium gas are enclosed in a vessel of volume V litre at TK. If V_i is volume available for expansion of helium, then the diameter of He atom is : (Assume real gas behavior for He)

$$A) \left(\frac{3V_i}{2\pi N_A n} \right)^{1/3} \quad B) \left(\frac{3(V - V_i)}{2\pi N_A n} \right)^{1/3} \quad C) \left(\frac{6(V - V_i)}{\pi N_A n} \right)^{1/3} \quad D) \left(\frac{6V_i}{\pi n N_A} \right)^{1/3}$$

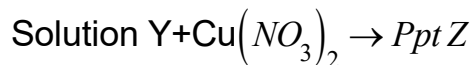
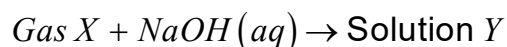
27. The density of the vapour of a substance at 1 atm pressure and 500 K is 0.36 kg m^{-3} . The vapour effuses through a small hole at a rate of 1.33 times faster than oxygen under the same condition. Which of the following is incorrect for the vapour?

- A) The molecular weight is 18
 B) It shows positive deviation under given conditions.
 C) It shows negative deviation under given conditions.
 D) Its compressibility is greater than one.

28. 100 ml of HCN is required for titration with 5M KMnO_4 solution in acidic medium. Side products are nitrate ion and carbondioxide along with manganese ion. CO_2 on treatment with slaked lime gives 10 gm of white ppt. Normality of HCN and volume KMnO_4 in ml is :

- A) 10 N, 40 ml B) 1 N, 8 ml C) 10 N, 20 ml D) 1 N, 2 ml

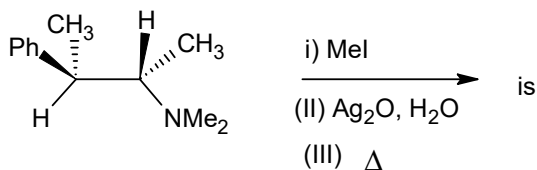
29. Consider the following reactions:

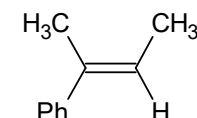
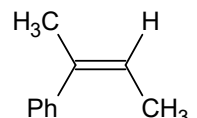
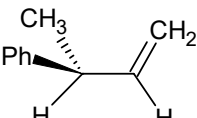
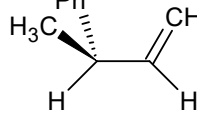


Ppt Z is

- A) Green coloured compound B) White coloured compound
 C) Yellow coloured compound D) Brown coloured compound

30. The major product obtained in the following reaction



- A)  B)  C)  D) 

SECTION - II

(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.
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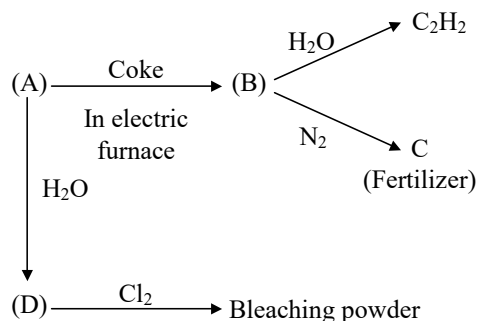
Paragraph for Question Nos. 31 & 32

A divalent metal hydroxide, $M(OH)_2$ is partly soluble in water. When the solid is shaken with water, most of it remain undissolved. However, the saturated solution obtained, was found to be basic and had a $pH = 10$.

In another experiment a soluble salt of M^{2+} was taken in a test tube and NH_4OH was added to it. Precipitate of $M(OH)_2$ was instantaneously formed. Under the circumstances, the precipitate was shaken with solid NH_4Cl . The precipitate dissolved.

31. What is the minimum concentration of OH^- , required for precipitation of $M(OH)_2$ in a solution containing $0.01M M^{2+}$ ions?
 A) $5 \times 10^{-11}M$ B) $2 \times 10^{-5}M$ C) $10^{-4}M$ D) $7.1 \times 10^{-6}M$
32. What should be the minimum concentration of NH_4Cl that must be present to prevent precipitation, when $0.01M NH_4OH$ is added to $0.01M M^{2+}$ solution?
 (K_b of $NH_4OH = 1.8 \times 10^{-5}$)
 A) $0.025M$ B) $0.01M$ C) $0.25M$ D) $0.005M$

Paragraph for Question Nos. 33 & 34



33. What is not true about A?
 A) It is flux, used in the metallurgy of Fe
 B) It is used for softening of water
 C) Its aqueous solution shows turbidity of yellow colour
 D) Addition of water to A is exothermic process.
34. Which statement is correct?
 A) $B \rightarrow C_2H_2$ is acid base reaction B) $B \rightarrow C_2H_2$ is redox reaction
 C) $C \xrightarrow{H_2O} HNO_3$ D) B is used as colour pigment

Paragraph for Question Nos. 35 & 36

By proposing stationary orbits Bohr explained the stability of the atom. He quantized the angular momentum of an electron in an orbit and calculated the energy, radius and

velocity of an electron in an orbit in one electron species. He considered energy changes takes place discontinuously and explained the line spectrum of one electron species.

35. The orbital angular momentum of electron in 2s orbital in hydrogen atom is
 A) 0 B) $\sqrt{2} \frac{h}{2\pi}$ C) $\sqrt{6} \frac{h}{2\pi}$ D) None of these
36. The excited electron in hydrogen atom jumped from fourth orbit to second orbit then the correct statement among the following is
 A) its total energy decreases by two times
 B) its kinetic energy increases by four times
 C) its potential energy decreases by two times
 D) its total energy increases by four times

SECTION - III
(Matching List Type)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

37. Match the following

Column-1	Column-2
A. Liquid sodium metal	p. Breathing apparatus submarine.
B. Potassium stearate	q. Explosive
C. Potassium nitrate	r. Coolant in nuclear reactor
D. Potassium superoxide	s. Soft Soap

A) A-p, B-q, C-r, D-s

B) A-r, B-s, C-p, D-q

C) A-r, B-s, C-q, D-p

D) A-r, B-q, C-s, D-p

38. Match the following

Column (I)

A) Ethene + $\text{Br}_2(\text{CCl}_4)$

B) Cis 2-butene + $\text{Br}_2(\text{CCl}_4)$

C) Trans 2-butene + $\text{OsO}_4/\text{NaHSO}_3$

D) Acidic hydration of propene

A) A-q, B-q, s, C-p, s, D-r

C) A-s, B-p, C-q, s, D-r

Column (II)

(p) Syn addition

(q) Anti addition reaction

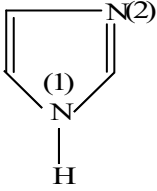
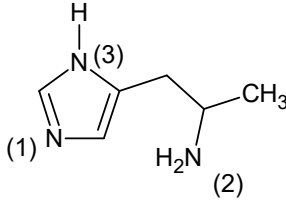
(r) Acyclic intermediate

(s) Racemic mixture

B) A-s, B-q, r, C-p, s, D-r

D) A-q, r, B-q, s, C-p, s, D-r

39. Match the following

Column – 1	Column – 2
A) $\begin{array}{c} 1 \qquad 2 \\ \text{NH}_2 - \text{C} - \text{NH}_2 \\ \parallel \\ \text{NH} \\ 3 \end{array}$	p. N-1 and N – 2 are equally basic
B) 	q. N – 1 is more basic than N – 2
C) 	r. N- 2 is more basic than N – 1
D) $\text{CH}_3 - \overset{(2)}{\underset{\text{O}}{\parallel}{\text{C}}} - \overset{(1)}{\text{NH}} - \text{CH}_2 - \text{CH}_2 - \text{NH}_2$	s. N – 3 is less basic than N – 1 and N – 2

A) A-p, s, B-q, C-r, s, D-p

B) A-p, s, B-q, C-p, D-r

C) A-p, B-r, C-r, s, D-r

D) A-p, B-r, C-r, s, D-q

40. Match the following :

	Column-I		Column-II
A)	4.5 m solution of H ₂ SO ₄ density 1.44 gm/ml	p)	Mole fraction of solute is 0.2 (approx)
B)	3 M 100 ml H ₂ SO ₄ mixed with 1 M 300 ml H ₂ SO ₄ solution	q)	Molarity 1.5
C)	14.5 m aq solution of urea	r)	Molarity = 4.5
D)	Into 4M two litre solution of NaOH, 40 gm NaOH is added.	s)	Mole fraction of solute is 0.35

A) A – p ; B – q ; C – r ; D – s,

B) A – r ; B – s ; C – p ; D – q

C) A – r ; B – q ; C – p ; D – r,

D) A – r ; B – q ; C – p ; D – s,

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

41. A line passes through (2, 0). Then which of the following is the slope of the line, for which its intercept between $y = x - 1$ and $y = -x + 1$ subtends a right angle at the origin
- A) $-\frac{1}{\sqrt{3}}$ B) $-\sqrt{3}$ C) $\sqrt{3}$ D) None of these
42. A, B are 5×5 matrices $\left(A = [a_{ij}]; B = [b_{ij}] \right)$ such that $a_{ij} = 0$ if $i \leq j$ and $b_{ij} = 0$ if $i \leq j + 1$.
- A) $A^5 = O$ B) $B^5 = O$ C) $A^4 = O$ D) $B^4 = O$
43. The volume of a right triangular prism $ABC A_1 B_1 C_1$ is equal to 3. Then the co-ordinates of the vertex A_1 , if the co-ordinates of the base vertices of the prism are $A(1,0,1)$, $B(2,0,0)$ and $C(0,1,0)$ be (a,b,c) then $a+b+c$ can be equal to (A_1 is vertically above A)
- A) 4 B) 0 C) -2 D) 2
44. If $\alpha(\bar{a} \times \bar{b}) + \beta(\bar{b} \times \bar{c}) + \gamma(\bar{c} \times \bar{a}) = \bar{0}$, Then
- A) $\bar{a}, \bar{b}, \bar{c}$ are coplanar only if none of α, β, γ is zero
- B) $\bar{a}, \bar{b}, \bar{c}$ are coplanar if atleast one of α, β, γ is non zero
- C) $\bar{a}, \bar{b}, \bar{c}$ are non-coplanar for any α, β, γ
- D) none of these
45. The value of $\lim_{x \rightarrow 0^+} \left(\ln(\sin^3 x) - \ln(x^4 + ex^3) \right) =$
- A) 0 B) 1 C) -1 D) does not exist
46. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the equation $f(x)f(y) - f(xy) = x + y, \forall x, y \in \mathbb{R}$ and $f(1) > 0$, then
- A) $f(x)f^{-1}(x) = x^2 - 4$ B) $f(x)f^{-1}(x) = x^2 - 6$
- C) $f(x)f^{-1}(x) = x^2 - 1$ D) none of these
47. $|x^2 \sin x + \cos^2 x e^x + \ln^2 x| < x^2 |\sin x| + \cos^2 x e^x + \ln^2 x$ true for $x \in$
- A) $(-\pi, 0)$ B) $\left(0, \frac{\pi}{2}\right)$ C) $(2n\pi, (2n+1)\pi) n \in \mathbb{N}$ D) None of these

48. M is the mid point of side AB of equilateral triangle ABC. P is a point on BC such that AP + PM is minimum. If AB = 20 then AP + PM is
 A) $10\sqrt{7}$ B) $10\sqrt{3}$ C) $10\sqrt{5}$ D) 10
49. If $f(2010) = \pi/2$ & $f(x) = f(2x) \forall x \in R$ & $f(x)$ is continuous for $x \in R$, then the value of $\lim_{x \rightarrow 0} \frac{\cos(f(x)) - \sqrt[3]{\cos f(x)}}{\sin^2 x}$
 A) 1/4 B) 1 C) 0 D) non-existent
50. If the pairs of lines $3x^2 - 2pxy - 3y^2 = 0$ and $5x^2 - 2qxy - 5y^2 = 0$ are such that each pair bisects the angle between the other pair, the pq equals to
 A) -1 B) -7 C) -9 D) -15

SECTION - II
(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 51 & 52

Let $A = [a_{ij}]$ be a square matrix of order 3 whose elements are distinct integers from 1, 2,.....,9, the matrix is formed so that the sum of numbers in every row, every column & diagonal is a multiple of 9.

51. The element a_{22} must be a multiple of
 A) 2 B) 3 C) 4 D) 9
52. The maximum value of trace of the matrix A is:
 A) 18 B) 19 C) 12 D) None

Paragraph for Question Nos. 53 & 54

The vertices of a triangle ABC are $A \equiv (2, 0, 2)$, $B \equiv (-1, 1, 1)$ and $C \equiv (1, -2, 4)$. the points D and E divide the side AB and CA in the ratio 1 : 2 respectively. Another point F is taken in space such that perpendicular drawn from F on ΔABC , meets the triangle at the point of intersection of the line segment CD and BE, say P. if the distance of F from the plane of the ΔABC is units $\sqrt{2}$, then

53. The position vector of P, is
 A) $\hat{i} + \hat{j} - 3\hat{k}$ B) $\hat{i} - \hat{j} + 3\hat{k}$ C) $2\hat{i} - \hat{j} - 3\hat{k}$ D) none
54. The equation of the line AF, is:
 A) $\vec{r} = (2\hat{i} + 2\hat{k}) + \lambda(\hat{i} + 2\hat{k})$ B) $\vec{r} = (2\hat{i} + \hat{k}) + \lambda(-\hat{i} + \hat{k})$
 C) $\vec{r} = (\hat{i} + \hat{k}) + \lambda(\hat{i} + 2\hat{k})$ D) $\vec{r} = (2\hat{i} + 2\hat{k}) + \lambda(-\hat{i} + 2\hat{k})$

Paragraph for Question Nos. 55 & 56

The graph of the function $f(x) = x^{10} + 9x^9 + 7x^8 + \dots + a_1x + a_0$ intersect the line $y = b$ at the points $B_1, B_2, B_3, \dots, B_{10}$ (from left to right) and the line $y = c$ at the points $C_1, C_2, C_3, \dots, C_{10}$ (from left to right). Let P be a point on the line $y = c$ to the right of the point C_{10} . If $b = 5$ and $c = 3$, then

55. Find sum of abscissas of points $B_1, B_2, B_3, \dots, B_{10}$

- A) 0 B) 9 C) -9 D) none of these

56. Find the sum $\sum_{n=1}^{10} \cot(B_n C_n P)$.

- A) 0 B) 1 C) 2 D) 4

SECTION-III**(MATCHING LIST TYPE)**

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

57. Match the general solution of the trigonometric equations given in Column-I with their corresponding entries given in Column-II.

Column-I	Column-II
P) $\cos^2 2x + \cos^2 x = 1$	1) $x = (n\pi + \frac{\pi}{4}) \cup (n\pi + \frac{\pi}{6}), n \in I$
Q) $\cos x = \sqrt{3}(1 - \sin x)$	2) $x = \frac{n\pi}{3}, n \in I$
R) $1 + \sqrt{3} \tan^2 x = (1 + \sqrt{3}) \tan x$	3) $x = (2n-1)\frac{\pi}{6}, n \in I$
S) $\tan 3x - \tan 2x - \tan x = 0$	4) $x = (2n\pi + \frac{\pi}{2}) \cup (2n\pi + \frac{\pi}{6}), n \in I$

Code :

P	Q	R	S
A) 1	2	3	4
B) 4	3	2	1
C) 2	1	3	4
D) 3	4	1	2

58. Consider, $f(x) = x - \sin x$, $g(x) = \cos^{-1}\left(e^{\frac{-x^4}{2}}\right)$ and $h(x) = \tan^{-1}\left(\frac{2x}{1+x^2}\right)$

	List -I		List -II
p)	$\lim_{x \rightarrow 0} \frac{f(x) - g(x)}{x^2}$ is equal to	1)	-1
q)	$\lim_{x \rightarrow 0} \frac{g(x) - x \cdot h(x)}{x^2}$ is equal to	2)	$-\frac{2}{3}$
r)	$\lim_{x \rightarrow 0} \frac{f(2x) - h(x^3)}{x^3}$ is equal to	3)	$-\frac{1}{2}$

s)	$\lim_{x \rightarrow 0} \frac{3f(x)}{x^3}$ is equal to	4)	$\frac{1}{2}$
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[Note: $[y]$ denotes greatest integer less than or equal to y]

A) P-1, Q-1, R-2, S-4

B) P-1, Q-2, R-1, S-1

C) P-1, Q-2, R-1, S-2

D) P-1, Q-1, R-2, S-1

59. Match the following.

	Column-I		Column-II
P)	The number of points of non-derivability of the function $f(x) = \left[\frac{2x}{\pi} \right] \operatorname{sgn} \left(\frac{1}{\{x\}} \right)$ in $(-2, 2)$ is (where $[y]$, $\{y\}$ and $\operatorname{sgn}(y)$ denote largest integer $\leq y$ fractional part of y and signum function of y respectively)	A	7
Q)	If $f: [0, \infty) \rightarrow R$ be a function defined as $f(x) = \begin{cases} 2 \sin \sqrt{x} + \sqrt{x} \sin \left(\frac{1}{x} \right)^x, & x > 0 \\ K, & x = 0 \end{cases}$ is continuous at $x=0$ then $7K$ is	B	3
R)	Let ABC be a variable triangle such that A is $(1, 2)$, B and C lie on the line $y = x + \lambda$ (where λ is a real variable except 1). The locus of the orthocenter of triangle ABC is a straight line whose y-intercept is equal to	C	4
S)	The number of integral values of x satisfying $\frac{\left(2^{\frac{\pi}{\tan^{-1} x}} - 4 \right) (x-4)(x-9)}{x!(x-1)!} < 0$, is	D	5

A) P-D, Q-A, R-B, S-C

B) P-A, Q-B, R-D, S-C

C) P-D, Q-C, R-B, S-A

D) P-C, Q-B, R-A, S-D

60. Match the following column-I with column- II.

	Column I		Column II
(P)	Foot of the perpendicular drawn from the point $(2, 3)$ to the line joining the points $(8, 10)$ and $(6, 10)$ is the point	1	$(14, 16)$
(Q)	Image of the point $(2, 4)$ in the line $x + y = 12$ is the point	2	$(-2, 0)$
(R)	Harmonic conjugate of the point $(6, 8)$ with respect to the point $(2, 4)$ and $(8, 10)$ is the point	3	$(2, 10)$

(S)	The points A and B are (2, 4) and (6, 0) respectively. If AB is rotated through an angle of 90° in the clockwise direction about the point A then the new position of B becomes the point	4	(8, 10)
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	P	Q	R	S		P	Q	R	S
A)	2	1	4	3	B)	3	4	1	2
C)	4	1	3	2	D)	3	1	2	4

KEY SHEET

1	C	2	C	3	B	4	B	5	C
6	C	7	B	8	C	9	A	10	A
11	B	12	A	13	B	14	A	15	B
16	B	17	B	18	A	19	D	20	D
21	D	22	D	23	C	24	B	25	B
26	B	27	C	28	A	29	A	30	B
31	D	32	A	33	C	34	A	35	A
36	B	37	C	38	A	39	D	40	C
41	A	42	A	43	C	44	B	45	C
46	C	47	D	48	A	49	C	50	D
51	B	52	A	53	B	54	D	55	C
56	A	57	D	58	A	59	A	60	B